Peaking of Fluid Displacement Drag

It was shown experimentally in references 10 and 15 that fluid displacement drag reaches a maximum at a ground speed near the tire hydroplaning speed (fig. 16). Recent data obtained at the Langley landing loads track (fig. 17(a)) illustrate this effect more clearly. Shown in this figure is the effect of tire inflation pressure on both fluid drag magnitude and peak location with ground speed. It can be seen from these data that increasing the ground speed above the critical hydroplaning speed results in appreciable reductions in fluid drag. This result is attributed to the tires lifting off the runway surface at the higher ground speeds and consequently displacing less runway fluid from the tire paths. The drag at speeds above V_P may be slightly in error because of the normalizing processes used.

Loss in Braking Traction

When rib tread tires are braked on most wet but not flooded pavement surfaces, the wet-runway friction coefficients obtained are usually considerably reduced in magnitude from the dry-runway values experienced but an appreciable amount of braking traction is still retained for this wet condition even at the highest vehicle ground speeds as shown in figure 18. On the other hand, when deep puddles form on the wet pavement surface, an intermittent additional loss in braking traction occurs because of hydroplaning of the tires in the puddles whenever the vehicle ground speed exceeds the tire hydroplaning speed as shown in figure 19. Whenever the pavement surface is flooded with fluids such as slush or water to depths large enough to initiate tire hydroplaning, the braking traction loss becomes catastrophic (braking friction coefficients approach free rolling friction coefficient) at ground speeds near or in excess of the tire hydroplaning speed. (See fig. 20.) This result is obvious when the unbraked-tire spin-down data under hydroplaning conditions previously discussed are considered. (See figs. 10, 11, and 12.) It is apparent from these data that applying brakes to wheels that have either completely or nearly stopped rotating from hydroplaning effects cannot be expected to improve the existing tire retardation forces and friction coefficient at all.

Loss of Directional Stability

Another significant manifestation of tire hydroplaning is a loss of directional stability, as demonstrated during the slush study conducted by the Federal Aviation Agency with technical assistance of the National Aeronautics and Space Administration on a four-engine jet transport. (See refs. 10, 15, and 17.) Several test runs through the slush bed were made at speeds in excess of the predicted hydroplaning speed for the airplane (110 knots) in the presence of direct cross-wind components that did not exceed 9 knots. The test aircraft at a ground speed of 120 knots was observed to yaw and drift laterally on the runway while in the slush bed. (These instances are shown in the motion pictures of ref. 17.) These results appear to indicate that loss of tire directional stability at and above tire hydroplaning speeds could be extremely serious to some aircraft when take-offs and landings are conducted in the presence of high cross winds.